

Math 421

Monday, November 10

Announcements

- Exam Week Drop-In Hours
 - Tuesday 1:30-3:30 pm VV B205
 - Wednesday 12:00-1:00 pm VV B205
- Exam 2
 - **Date:** Wednesday, 11/12
 - **Time:** 5:45-7:15 pm
 - **Location:** Birge 145

Continuity – Equivalent Definitions

Theorem. Let $f: A \rightarrow \mathbb{R}$ and let $x_0 \in A$. Then the following are equivalent:

a) f is continuous at x_0 ,

b) via sequences:

c) via $\epsilon - \delta$:

d) $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Activity – Prove Using $\epsilon - \delta$ Definition

1. If $a \in \mathbb{R}$ and $g(x) = a$ is the constant function, then g is continuous at all $c \in \mathbb{R}$.
2. If $f(x) = x$, then f is continuous at all $c \in \mathbb{R}$.

Algebraic Continuity Properties

Theorem 17.3 & 17.4. Assume $f: A \rightarrow \mathbb{R}$ and $g: A \rightarrow \mathbb{R}$ are both continuous at a point $x_0 \in A$. Then

1. $|f(x)|$ is continuous at x_0 ,
2. $kf(x)$ is continuous at x_0 for all $k \in \mathbb{R}$,
3. $f(x) + g(x)$ is continuous at x_0 ,
4. $f(x)g(x)$ is continuous at x_0 , and
5. If $g(x_0) \neq 0$, then $\frac{f(x)}{g(x)}$ is continuous at x_0 .

Example

1. If P is a polynomial, then P is continuous at all $c \in \mathbb{R}$
2. Define $f: [0, \infty) \rightarrow [0, \infty)$ by $f(x) = \sqrt{x}$. Then f is continuous at all $c \in [0, \infty)$.

Composition of Continuous Functions

Theorem 17.5. Suppose $f: A \rightarrow \mathbb{R}$ and $g: B \rightarrow \mathbb{R}$. Assume $\{f(x): x \in A\} \subseteq B$ so that the composition $(g \circ f)(x) = g(f(x))$ is defined on A .

If f is continuous at $x_0 \in A$ and g is continuous at $f(x_0) \in B$, then the $g \circ f$ is continuous at x_0 .

Example

$$\text{Let } f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Example – Popcorn Function

$$\text{Let } f(x) = \begin{cases} \frac{1}{q}, & x \in \mathbb{Q} \text{ and } x = \frac{p}{q} \text{ in lowest terms} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Chapter 3: Continuity

Section 18: Properties of Continuous Functions

Bounded Functions

Definition: A real-valued function f is said to be **bounded** if

- ❖ Give an example of a function that is bounded on $\mathbb{R}$ and a function that is unbounded on $\mathbb{R}$.

Extreme Value Theorem

Theorem. Let f be a continuous real-valued function on a closed interval $[a, b]$. Then

- f is a bounded function.
- f assumes its maximum and minimum values on $[a, b]$. That is, there exist x_0, y_0 in $[a, b]$ such that $f(x_0) \leq f(x) \leq f(y_0)$ for all $x \in [a, b]$.

Discussion Questions

- ❖ Remove the continuity restriction on f from the hypotheses of the theorem. Does the conclusion still hold? If not, find a counterexample.
- ❖ Replace the closed interval with an open interval. Does the conclusion still hold? If not, find a counterexample.

Proof Idea

Boundedness:

Proof Idea

Maximum Attained:

Intermediate Value Theorem

Thm. If f is a continuous real-valued function on an interval I , then f has the intermediate value property on I :

Proof Idea

Example

Let $f: [0,1] \rightarrow [0,1]$ be a continuous function. Show that f has a fixed point, that is, there exists $x_0 \in [0,1]$ such that $f(x_0) = x_0$.